

Passive Radiative Cooling via Biomimetic Metasurfaces on Maritime Infrastructure

A Distributed Approach to Ocean Thermal Mitigation

Register 1 Blueprint Category: Technomouse and the Genie of the Bottle

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Abstract

Thermal expansion driven by global climate shifts contributes significantly to sea-level rise. While large-scale geoengineering poses severe ecological risks, localized, passive thermodynamic interventions offer a scalable alternative. This report details the theoretical application of highly emissive metasurfaces to existing maritime infrastructure. By utilizing the decks and containers of the global shipping fleet as a distributed flotilla of “Nautilus” nodes, this framework demonstrates a zero-energy mechanism to reject thermal energy through the $8 - 13 \mu\text{m}$ atmospheric window directly into deep space.

1 Introduction

The rise in global sea levels is driven primarily by the melting of terrestrial ice and the thermal expansion of seawater. The volumetric expansion of the upper oceanic layers accounts for a significant proportion (approximately 50 – 57%) of observed sea-level rise. Addressing this requires actionable, scalable interventions—a pragmatic architectural methodology that prioritizes functional, localized prototypes over unwieldy global mega-projects. The “Nautilus” concept proposes retrofitting existing oceanic assets to serve as passive thermal regulators, operating independently of centralized power grids.

2 Physical and Mathematical Framework

2.1 Volume Reduction via Thermal Exchange

The relationship between the volume V of a water mass and its temperature T is governed by the volumetric thermal expansion coefficient β . The change in volume ΔV

resulting from a localized temperature drop ΔT is approximated as:

$$\Delta V = \beta V_0 \Delta T \quad (1)$$

By engineering a local temperature reduction, the physical column of water undergoes a volumetric contraction, creating a micro-lowering effect in the immediate environment.

2.2 Net Radiative Cooling Power

To achieve this cooling without mechanical pumps or moving parts, the system relies entirely on passive daytime radiative cooling (PDRC). The net cooling power P_{cool} of a surface exposed to the sky is determined by the balance of outgoing thermal radiation and incoming heat fluxes:

$$P_{cool} = P_{rad} - P_{atm} - P_{sun} - P_{cond/conv} \quad (2)$$

Where:

- P_{rad} is the power radiated by the surface.
- P_{atm} is the absorbed atmospheric radiation.
- P_{sun} is the incident solar radiation.
- $P_{cond/conv}$ encompasses conductive and convective heat gains.

The outward radiation power P_{rad} depends on the surface emissivity ϵ and temperature T , integrating over the hemispherical solid angle Ω and wavelength λ :

$$P_{rad} = A \int d\Omega \cos \theta \int_0^\infty d\lambda I_{BB}(T, \lambda) \epsilon(\lambda, \theta) \quad (3)$$

Here, $I_{BB}(T, \lambda)$ represents the spectral radiance of a blackbody at temperature T . To maximize P_{cool} during daylight, the surface must exhibit high reflectivity in the solar spectrum ($0.3 - 2.5 \mu m$) to minimize P_{sun} , while maintaining near-unity emissivity ϵ within the primary atmospheric transparency window ($8 - 13 \mu m$).

3 Metasurface Implementation

Optimized geometries for directional thermal management can be extracted from structural principles rather than biological systems, ensuring universal applicability and extreme environmental durability. By utilizing a precise triangular cross-section geometry that induces total internal reflection of solar radiation while maximizing emissivity in the mid-infrared range, a durable synthetic metasurface can be manufactured.

Nano-imprinting this geometric configuration onto fluoropolymer matrices or silica-based films yields a synthetic skin applicable to commercial maritime assets. Standard shipping containers and vessel superstructures—currently acting as thermal absorbers—would be retrofitted to act as passive heat-sinks. This angular selectivity ensures thermal emissions are directed orthogonally upward, bypassing Lambertian scattering back into the oceanic environment.

4 Strategic Deployment & Pitch Avenues

To operationalize this architecture, strategic alignment with existing maritime and engineering frameworks is required. The deployment roadmap targets three primary vectors:

- **Advanced Research Agencies:** Programs such as the ARPA-E or the NSF Convergence Accelerator represent primary funding avenues for the materials science validation of the metasurface. The mathematical framework proves viability without requiring massive infrastructure capital.
- **Commercial Maritime Conglomerates:** Application of the skin serves dual purposes. It functions as a hull and deck coating that reduces shipboard HVAC loads and minimizes drag and bio-fouling. Corporate venture arms can be pitched on the immediate fuel-efficiency ROI, with planetary cooling acting as a highly visible ESG (Environmental, Social, and Governance) byproduct.
- **Pre-Print Registries:** Archiving this mathematical framework in open-science repositories secures priority and integrates the localized cooling strategy into the global academic domain.

5 Conclusion

Integrating passive, structurally-derived metasurfaces onto existing fleets removes the barrier of deploying dedicated offshore structures. It transforms maritime infrastructure into a distributed, autonomous cooling network. By bridging theoretical thermodynamics with immediate, actionable maritime efficiency, this framework provides a highly viable methodology for localized oceanic thermal mitigation.

References

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